

Eenheid 1.9

Note Title

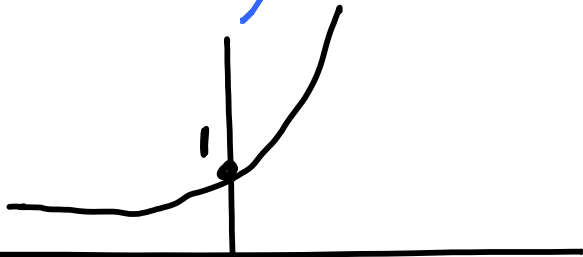
2016/02/12

Logaritmische functie

Stewart bl 59-62
Studiegids bl 12

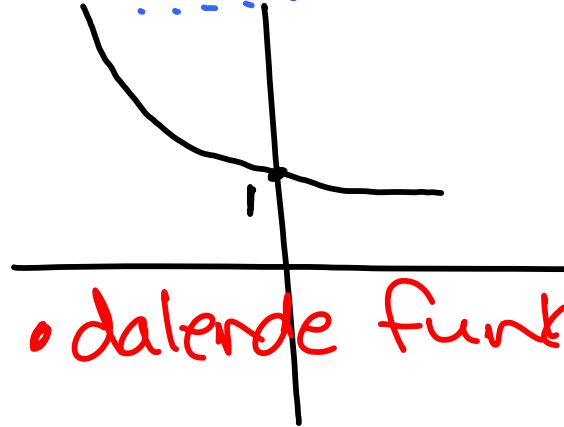
Bekou $y = a^x$

- $a > 1, a \neq 1$



• stijgende functie

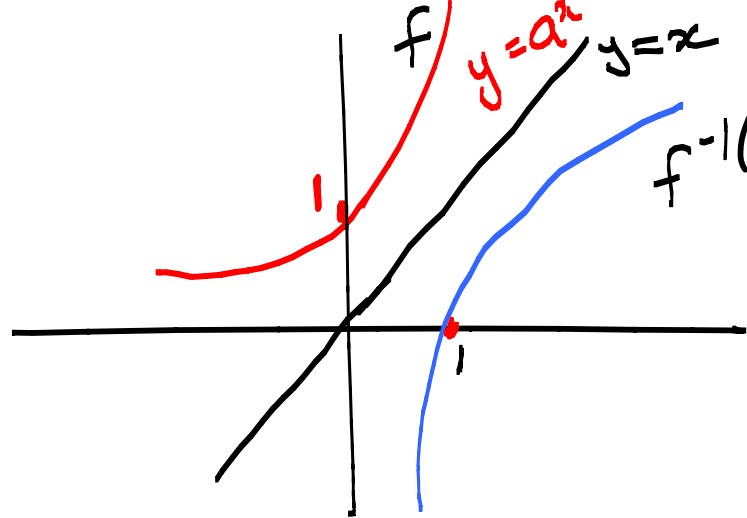
• $0 < a < 1$ Bv $(\frac{1}{2})^x = 2^{-x}$



• dalende functie

• Toepassing van horizontale lyn toets:

- functie net stijgend / net dalend
- f is dan een-tot-een
- f^{-1} bestaan



$$f^{-1}(x) = \log_a x$$

*Regstelling

Def. $x = a^y \Leftrightarrow y = \log_a x$

$$\text{III } y = a^x \Leftrightarrow x = a^y$$

$$\Leftrightarrow \log_a x = \log_a a^y$$

$$\Leftrightarrow \log_a x = y \log_a a = 1$$

$$\Leftrightarrow \log_a x = y$$

Bepaal x as $\log_2 8 = x$

Def: $\log_2 8 = x$

$$\Rightarrow 2^x = 8 = 2^3$$

$$\Rightarrow x = 3$$

of

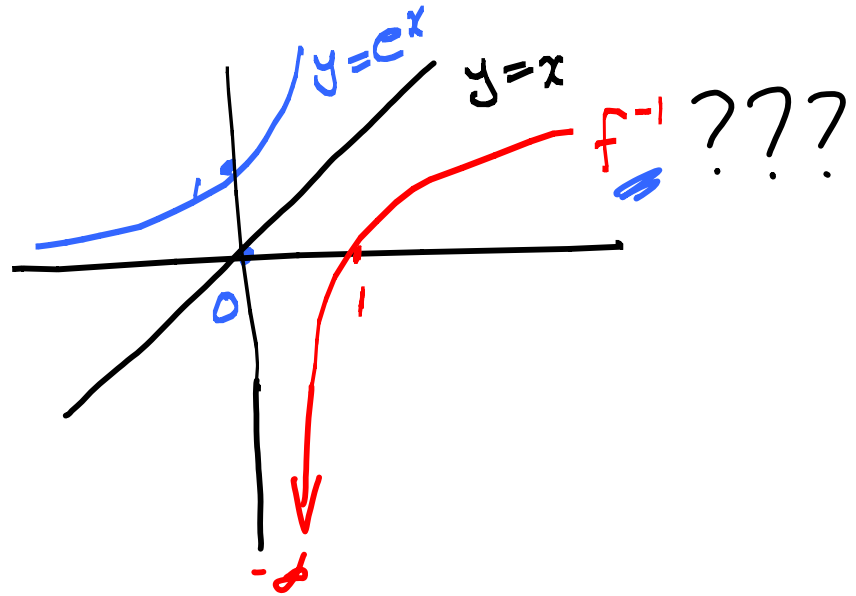
$$\log_2 8 = x \log_2 2 = 1$$

$$\Rightarrow \log_2 8 = \log_2 2^x$$

$$\Rightarrow 2^3 = 2^x = x = 3$$

Natuurlijke logaritmische functie bl 60

- $y = e^x \rightarrow$
- $f: \mathbb{R} \rightarrow (0, \infty)$
- f is een-tot-een
- f is stijgend



Dan $f^{-1}(x)$ bestaat

$$f^{-1}: (0, \infty) \rightarrow \mathbb{R}$$

$$* f^{-1}(x) = \boxed{\log_e} x = \boxed{\ln} x$$

• Opmerking: • $\ln e = 1$

Definition: $\bullet y = \ln x \Leftrightarrow x = e^y$

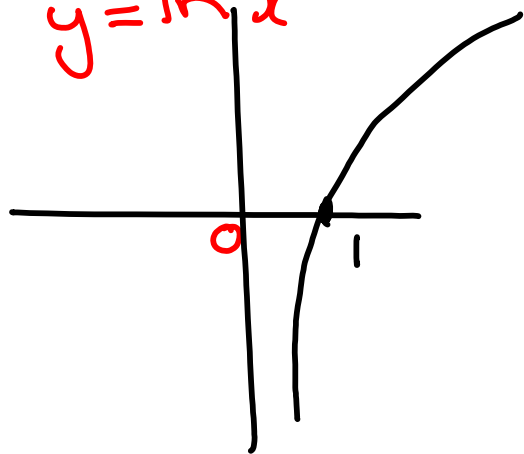
$$\bullet y = e^x \Leftrightarrow x = e^y$$

$$\Leftrightarrow \ln x = \ln e^y$$

$$\Leftrightarrow \ln x = y \ln e = y \quad (\Rightarrow \ln x = y)$$

* Sketch: $y = -\ln(x-2)$

$$y = \ln x$$



$$\bullet \ln 1 = 0$$

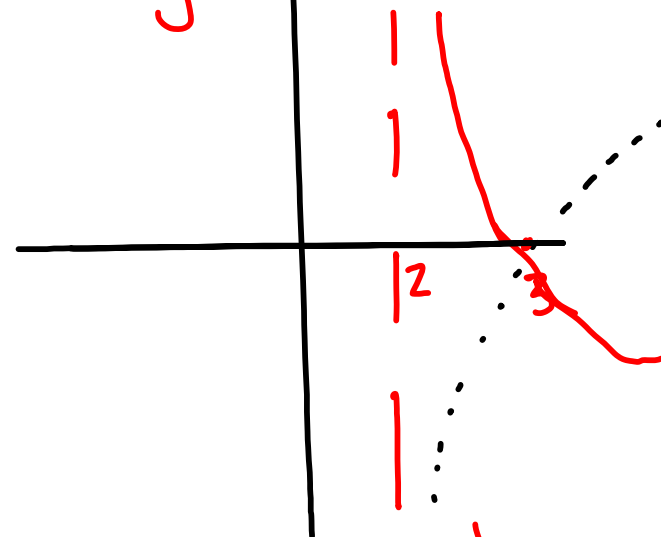
$$y = \ln(x-2)$$



$$\ln \boxed{\frac{(3-2)}{?}} = 0 \Rightarrow \ln 1 = 0$$

Orthon $y = \ln x \Rightarrow x > 0$

$$\bullet y = -\ln(x-2)$$

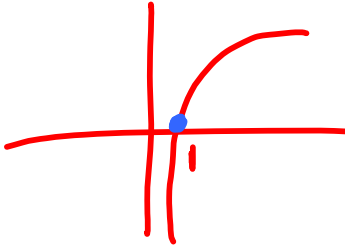


$$D_f = (2, \infty)$$

Eienskappe van ln-funksie

- ① $\ln(ab) = \ln a + \ln b$ $a, b > 0$
- ② $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$ $a > 0, b > 0$
- ③ $\ln(a^b) = b \ln a$ $a > 0$

NB • $y = \ln x$ ⁷⁰



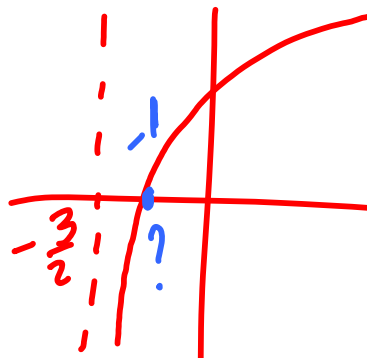
$$D_f = (0, \infty)$$
$$W_f = \mathbb{R}$$

• $g(x) = \ln(3 + 2x)$ ⁷⁰

$$3 + 2x > 0$$
$$2x > -3$$
$$x > -\frac{3}{2}$$

$$D_g = \left(-\frac{3}{2}, \infty\right)$$

$$W_g = \mathbb{R}$$



$\ln 1 = 0$
 $3 + 2x = 1$
 $2x = -2$
 $x = -1$

Kansellasielenskappe/vergelijking p61

- $f(x) = e^x$

$$D_f = \mathbb{R}, W_f = (0, \infty)$$

- $f^{-1}(x) = \ln x$

$$* D_{f^{-1}} = (0, \infty)$$

$$W_{f^{-1}} = \mathbb{R}$$

$$\begin{aligned} 1) (f^{-1} \circ f)(x) &= f^{-1}(f(x)) \rightarrow \boxed{x \in \mathbb{R} = D_f} \\ &= f^{-1}(e^x) = \underline{\ln e^x} = x \ln e = x \end{aligned}$$

- $\boxed{\underline{\ln} e^x = x \text{ vir } x \in \mathbb{R}}$

$$\begin{aligned} 2) (f \circ f^{-1})x &= f(f^{-1}(x)) \rightarrow x \in D_{f^{-1}} = (0, \infty) \\ &= f(\ln x) = \underline{e^{\ln x}} = e^y = x \end{aligned}$$

- $\boxed{\underline{e}^{\ln x} = x \text{ vir } x > 0 \text{ of } x \in (0, \infty)}$

Def:

$$\boxed{y = \ln x} \Rightarrow e^y = x$$

Los op : x? Gelykheid

of $\begin{cases} \text{kansellatie} \\ \text{eigenschap} \\ \text{Definitie} \end{cases}$

① $e^{5x} = 4$

kansellatie: $\ln e^{5x} = \ln 4$

$$\Rightarrow 5x \ln e = \ln 4$$

$$\Rightarrow 5x = \ln 4$$
$$x = \frac{1}{5} \ln 4$$

Def $e^{5x} = 4$

$$\Rightarrow 5x = \ln 4$$

$$\Rightarrow x = \frac{\ln 4}{5}$$

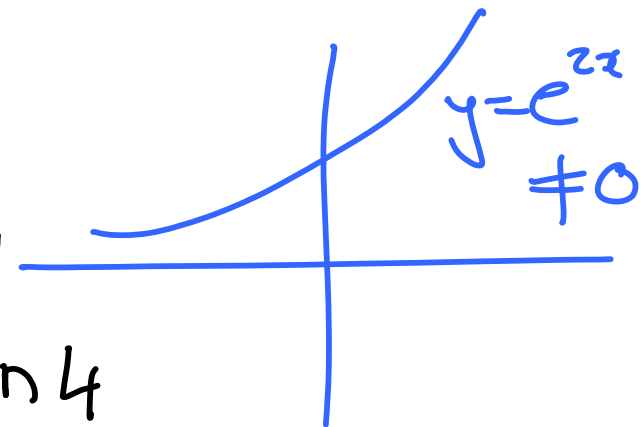
② $e^{5x} = 4 e^{2x}$

$$\Rightarrow \frac{e^{5x}}{e^{2x}} = 4 \Rightarrow e^{5x-2x} = 4$$

$$\Rightarrow \ln e^{3x} = \ln 4 \Rightarrow 3x \ln e = \ln 4$$

$$\Rightarrow x = \frac{1}{3} \ln 4$$

$$\boxed{e} = \boxed{k}$$



③ $e^{\ln 2x} = 8$
 $\Rightarrow 2x = e^8$
 $\Rightarrow x = \frac{e^8}{2}$

kansellasi

Def. $\ln 2x = 8$
 $\Rightarrow e^8 = 2x$
 $\Rightarrow x = \frac{e^8}{2}$

④ $\ln 3x = 8 + 2\ln x$

NB $\boxed{\ln} = \boxed{\ln}$

$\Rightarrow \ln 3x - 2\ln x = 8$

$\Rightarrow \ln 3x - \ln x^2 = 8$

$\ln p - \ln q = \ln \frac{p}{q}$

$\Rightarrow \ln \frac{3x}{x^2} = 8$

$\Rightarrow e^{\ln \frac{3}{x}} = e^8$

$\Rightarrow \frac{3}{x} = e^8$

$\Rightarrow x = \frac{3}{e^8}$

$$\textcircled{5} \quad e^x = 2e^2 \Rightarrow \frac{e^x}{e^2} = 2$$

$$\Rightarrow \ln e^{x-2} = \ln 2$$

$$\Rightarrow (x-2) \ln e^{\overset{=1}{} } = \ln 2 \Rightarrow x = \ln 2 + 2$$

$$\textcircled{6} \quad \ln x^2 - 6 = 0$$

$$\Rightarrow 2 \ln x = 6$$

$$\Rightarrow \ln x = \frac{6}{2} = 3$$

$$\Rightarrow e^{\ln x} = e^3 \Rightarrow x = e^3$$

$$\boxed{\ln} = \boxed{k}$$

$$\textcircled{7} \quad \ln 2x = 4 + 3 \ln x$$

$$\Rightarrow \ln 2x - \textcircled{3} \ln x = 4$$

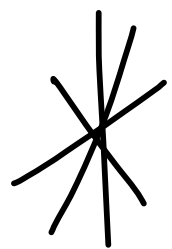
$$\Rightarrow \ln 2x - \ln x^3 = 4$$

$$\Rightarrow \ln \frac{2x}{x^3} = 4$$

$$\Rightarrow \textcircled{e} \ln \frac{2}{x^2} = \textcircled{4}$$

$$\Rightarrow \frac{2}{x^2} = e^4 \Rightarrow x^2 = \frac{2}{e^4}$$

$$\Rightarrow x = \frac{\sqrt{2}}{e^2}$$



$$e^{\ln x} = x \quad \text{vir } x > 0$$

8) Als $f(x) = 1 + e^{-2x}$

• bepaal $f^{-1}(x)$ en • skets $y = f(x)$ en $y = f^{-1}(x)$

$f(x) =$
• $y = 1 + e^{-2x}$
 $x = 1 + e^{-2y}$

$$\Rightarrow e^{-2y} = x - 1$$

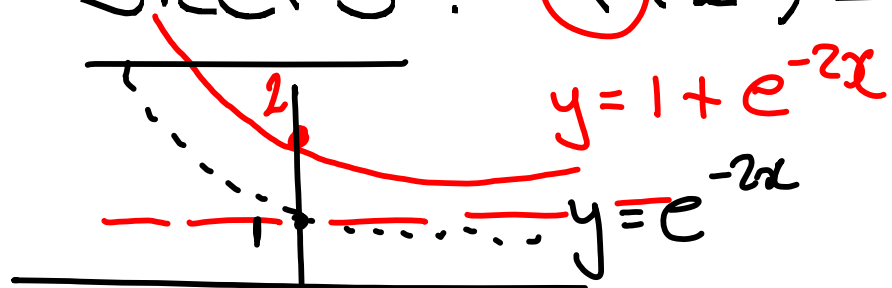
$$\Rightarrow -2y = \ln(x - 1)$$

$$\Rightarrow y = -\frac{1}{2} \ln(x - 1)$$

$$f(x) = 1 + e^{-2x}$$

$$f^{-1}(x) = -\frac{1}{2} \ln(x - 1)$$

Sketchs: $f(x) = 1 + e^{-2x}$

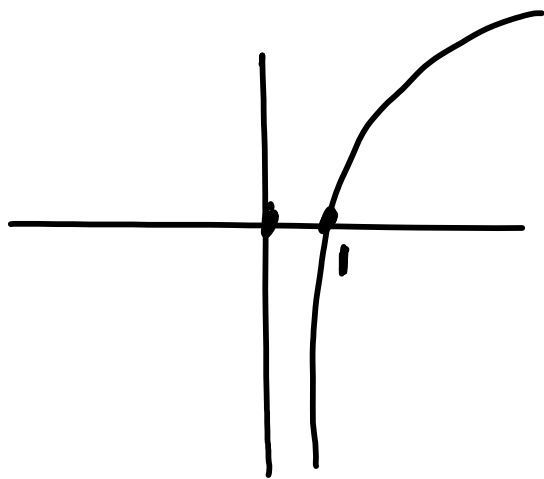


$$D_f = \mathbb{R}$$

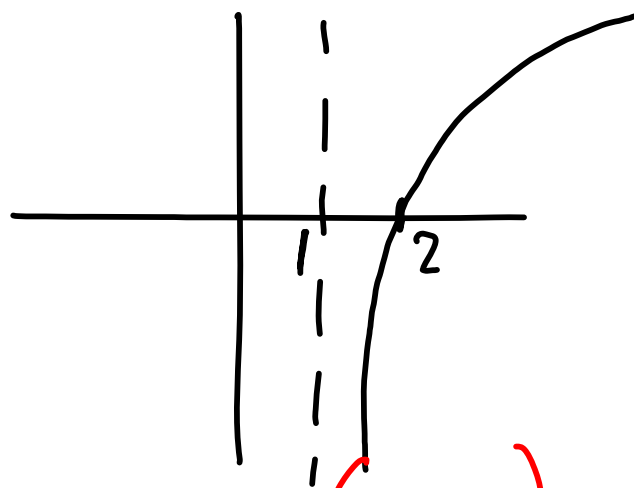
$$W_f = (1, \infty)$$

Sketchs: $y = f^{-1}(x) = -\frac{1}{2} \ln(x-1)$

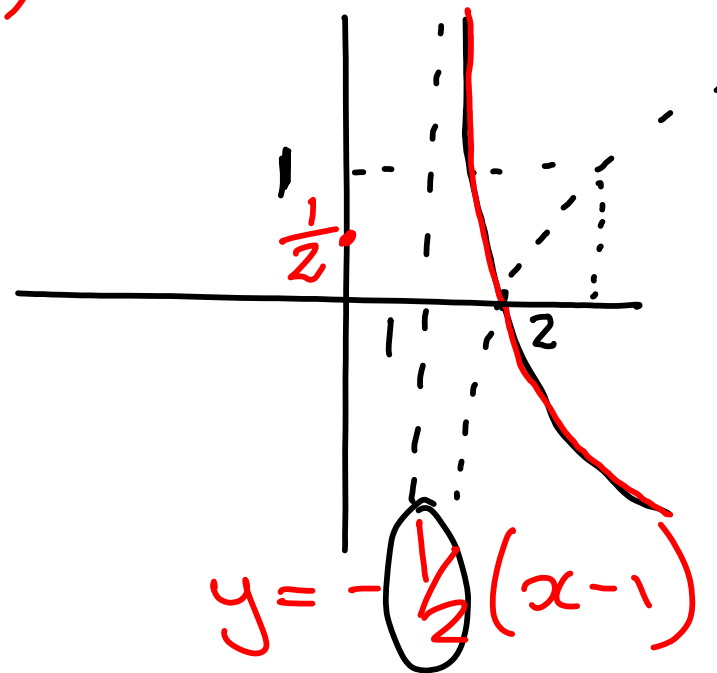
$$x=2 \Rightarrow \ln(2-1) = \ln 1 = 0$$



$$y = \ln x$$



$$y = \ln(x-1)$$



$$y = -\frac{1}{2} \ln(x-1)$$

$$W_f = D_{f^{-1}} = (1, \infty)$$

$$D_f = W_{f^{-1}} = \mathbb{R}$$

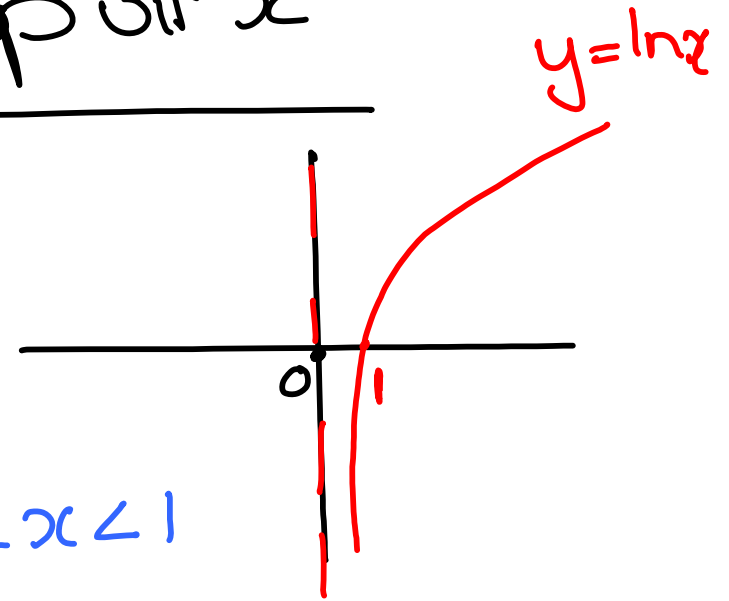
N3 Ongelykhede: Los op vir x

$$\textcircled{1} y = \ln x < 0$$

$$\Rightarrow 0 < x < 1$$

$$e^{\ln x} < e^0 \Rightarrow x < 1 \therefore 0 < x < 1$$

$$\boxed{\begin{array}{l} y = \ln x \\ x > 0 \end{array}}$$



$$\textcircled{2} e^x > 5$$

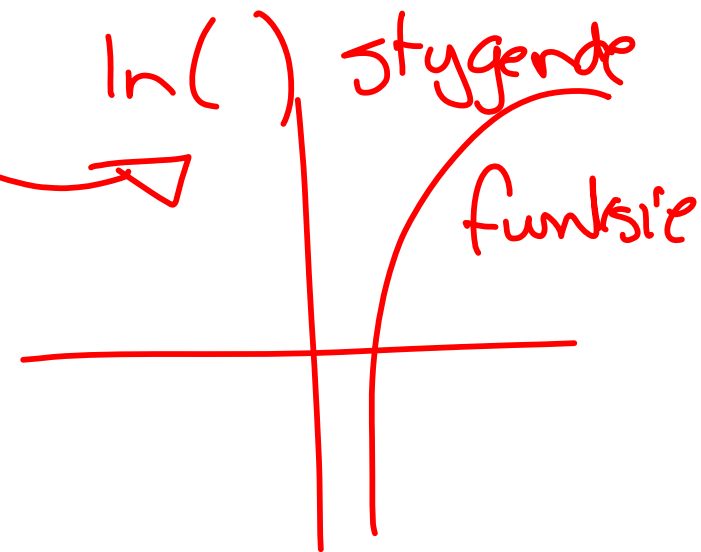
$$\Rightarrow \ln e^x > \ln 5$$

$$\Rightarrow x \ln e > \ln 5$$

$$\Rightarrow x > \ln 5$$

$$\Rightarrow x \in (\ln 5, \infty) \checkmark$$

$$\boxed{\begin{array}{l} y = e^x \\ x \in \mathbb{R} \end{array}}$$



$$\textcircled{3} \quad 1 < e^{3x-1} < 2$$

$$\begin{array}{l} y = e^{3x-1} \\ x \in \mathbb{R} \end{array}$$

$$\Rightarrow \ln 1 < \ln e^{3x-1} < \ln 2$$

(ln is 'n stygende funksie)

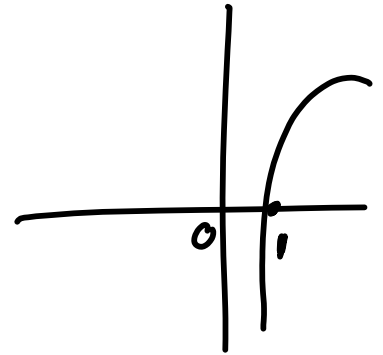
$$\Rightarrow 0 < (3x-1) \ln e < \ln 2$$

$$\Rightarrow 0 < 3x-1 < \ln 2$$

$$\Rightarrow 1 < 3x < 1 + \ln 2$$

$$\Rightarrow \frac{1}{3} < x < \frac{1 + \ln 2}{3}$$

$$\Rightarrow x \in \left(\frac{1}{3}, \frac{1 + \ln 2}{3} \right)$$



$$\textcircled{4} \quad \textcircled{1} - 2 \ln x < \textcircled{3}$$

$$\begin{aligned} y &= \ln x \\ \Rightarrow x &> 0 \\ &\dots \end{aligned}$$

$$\Rightarrow -2 \ln x < 2$$

$$\Rightarrow 2 \ln x > -2$$

$$\Rightarrow \ln x > -1 \quad (e^x \text{ is a strictly increasing function})$$

$$\Rightarrow x > e^{-1}$$

$$\Rightarrow x \in \left(\frac{1}{e}, \infty \right)$$

$$e^{-1} = \frac{1}{e}$$

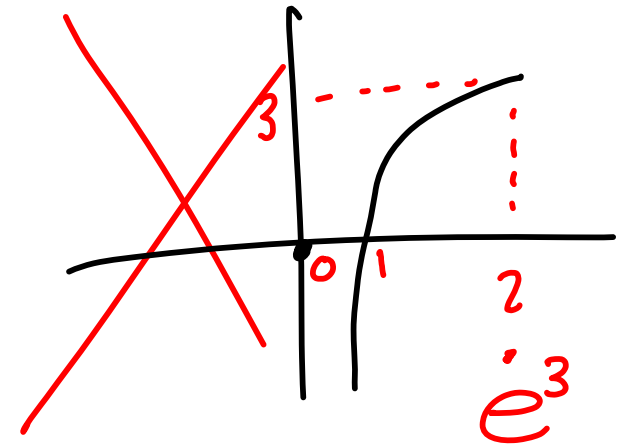
$$⑤ \quad y = \ln x < 3$$

$$\boxed{\begin{array}{l} y = \ln x \\ \Rightarrow x > 0 \end{array}}$$

$$\Rightarrow e^{\ln x} < e^3$$

$$\Rightarrow \underline{\underline{x < e^3}} \quad ?? \text{ en } x > 0$$

$$\Rightarrow x \in (0, e^3)$$



$$⑥ \quad f(x) = \ln(e^x - 3)$$

Bepaal • D_f • $f'(x)$ en • $D_{f^{-1}}$

• D_f

$$e^x - 3 > 0$$



